

Old Direction vs. New Direction

What changed, why, and where your original contribution lives
Ramon Gregorio — April 2026

1. Side-by-Side Comparison

Central question	Can a discrete 3-layer model reproduce Apollo HFE data?	What are the subsurface thermal profiles at 20 m resolution for Artemis south pole sites, and how do they affect ice stability and THz observations?
Regolith model	Discrete 3-layer: fixed ρ jumps at 2 cm and 20 cm	Continuous H-parameter (Hayne 2017) + Martinez-Siegler low-T correction + novel ice-coupled properties
Validation	Apollo 15/17 HFE (equatorial sites)	Diviner polar bolometric T (primary) + Apollo as sanity check
Spatial scope	Two equatorial point locations	20 m/pixel maps over 5×5 km patches at 3–5 south pole sites
Shadow modeling	80 m DEM, 72 azimuths, direct only	20 m DEM, 720 azimuths, direct + scattered visible + thermal IR via view factors
Downstream	Drill-bit thermal simulation (engineering)	TSUKIMI RTM coupling via Jacobian + ice survivability maps
Novel contribution	Discrete layer comparison with continuous model	Ice-coupled thermal properties (new physics), conductivity model sensitivity map, THz Jacobian analysis
Paper framing	Methods paper: 'Here is our model and it works'	Results paper: 'Here is what the model reveals about ice stability and what TSUKIMI will observe'
Target journal	Icarus	Planetary Science Journal (open-access, Artemis-relevant)
Collaboration	Solo (Kasai Lab)	TSUKIMI multi-institution: Tohoku, NICT, UTokyo, IST

2. Why the Direction Changed

Three things converged to force this pivot:

Your colleague's feedback was correct. The discrete 3-layer model with sudden density jumps at 2 cm and 20 cm does not represent real lunar regolith. Under self-weight compaction, density increases continuously with depth. The 3-layer model can be tuned to match Apollo data (any model with enough free parameters can), but it produces unphysical thermal gradients in the top 20 cm — exactly where the TSUKIMI terahertz radiometer is sensitive. A model that fits the data for the wrong physical reasons cannot be trusted for predictions at new locations.

TSUKIMI needs a pipeline, not a paper. Your primary deliverable is no longer a journal article about model validation — it's a functional pipeline that produces $T(z, t)$ grids your colleagues can plug into their RTM. This changes everything about how you prioritize work: spatial coverage matters more than exhaustive Apollo comparison, physical realism matters more than empirical curve-fitting, and computational efficiency matters because you need to run 250,000 pixels, not two.

The field moved. Between 2020 and 2025, new in situ data (Chang'E-4, ChaSTE, LISTER), new property models (Martinez-Siegler, Bürger), and new computational tools (HODLR view factors, 5 m DEMs) made it possible to do thermal modeling at a level that wasn't feasible when you started. Your thesis should reflect the current state of the art, not 2017 methods.

What stays the same: Your Crank-Nicolson solver, Numba acceleration, energy-balance boundary condition, ray-tracing algorithm, package structure, and scientific writing skills all carry forward unchanged. The pivot is in the property model (~30 lines of code), the spatial scope (loop over pixels instead of running two sites), and the framing (results paper instead of methods paper).

3. Your Novel Physics Contribution: Ice-Coupled Thermal Properties

You asked: is it just Hayne's work that would be used? **No**. Here is where you go beyond Hayne and add genuinely new physics to the 1D heat equation.

3.1 – The problem nobody has solved

Every published lunar thermal model (Hayne, Martinez-Siegler, Bürger, Paige, Vasavada) treats the regolith as **always dry**. They compute $T(z, t)$, then *separately* check if ice would be stable at those temperatures. This is a one-way coupling: thermal model → ice stability.

But if ice IS present, it changes the thermal properties of the regolith:

Property	Dry regolith	Ice at 100 K	Ratio
Thermal conductivity K	$\sim 10^{-3}$ W/m/K	~ 5.7 W/m/K	$\sim 5,000\times$
Specific heat c_p	~ 200 J/(kg·K)	~ 800 J/(kg·K)	$\sim 4\times$
Density ρ	~ 1500 kg/m ³	~ 917 kg/m ³	Pore-filling

Even **1–5 wt% ice** in the pore space dramatically increases the effective thermal conductivity because ice conducts heat $\sim 5,000\times$ better than vacuum-separated regolith grains. This means an icy layer conducts heat downward more efficiently, changing the entire subsurface thermal profile — which in turn changes where ice is stable.

3.2 – The feedback loop (why this is novel)

The ice-thermal feedback loop:

1. Start with the dry Hayne model → compute $T(z, t)$
2. From $T(z, t)$, compute ice stability depth z^* using Schorghofer-Williams sublimation rates
3. At depths where ice is stable ($z > z^*$), modify the thermal properties:
$$K_{\text{eff}}(z) = K_{\text{dry}}(z) + \phi_{\text{ice}}(z) \cdot K_{\text{ice}}(T)$$
$$c_{p,\text{eff}}(z) = (1 - \phi_{\text{ice}}) \cdot c_{p,\text{reg}}(T) + \phi_{\text{ice}} \cdot c_{p,\text{ice}}(T)$$
$$\rho_{\text{eff}}(z) = \rho_{\text{dry}}(z) + \phi_{\text{ice}}(z) \cdot \rho_{\text{ice}}$$
4. Re-run the thermal solver with updated properties → get new $T(z, t)$
5. Recompute ice stability → z^* may have shifted
6. Iterate until z^* converges (typically 3–5 iterations)

The key finding you'll produce: The dry-model ice stability depth z^*_{dry} differs from the self-consistent z^*_{coupled} . The sign of the shift tells you whether ice helps itself survive (positive feedback → ice expands the stable zone) or undermines its own stability (negative feedback). This has not been quantified for the Moon.

3.3 – Why nobody has done this yet

Three reasons: (1) Most thermal modelers (Hayne, Paige, Vasavada) work on global or mid-latitude problems where there is no ice, so the coupling is irrelevant. (2) Ice stability modelers (Schorghofer, Williams) treat the thermal profile as given and compute ice stability as a post-processing step. (3) The coupling requires iterating between two models, which adds computational cost and implementation complexity that nobody has invested in. But for your pipeline — which already computes both thermal profiles and ice stability —

adding the feedback loop is natural. It's maybe 50 lines of extra code wrapping your existing solver in an iteration loop.

3.4 – Connection to THz observations

This is where it gets powerful. If ice at 5–20 cm depth increases the local thermal conductivity by even 10× (which just 2% ice by volume would do), it changes the thermal gradient in the THz-sensitive zone. Your Jacobian analysis can then answer: *can TSUKIMI's 480 GHz radiometer distinguish an icy subsurface from a dry one based on the thermal gradient signature?* If yes, you've identified a new detection method for subsurface ice using terahertz radiometry — which is exactly what TSUKIMI was designed to investigate.

4. Other Possible Novel Contributions (Beyond Hayne)

The ice-coupled model is the strongest option, but here are other ways you could contribute genuinely new physics:

Depth-varying H-parameter

Hayne assumes H is constant with depth. But impact gardening, space weathering, and regolith maturation create a depth-dependent compaction history. You could model $H(z) = H_0 \cdot [1 + \alpha \cdot \ln(1 + z/z_0)]$ to capture the fact that deeper regolith has experienced more compaction.

Implementation difficulty: Moderate. Requires choosing functional form and justifying parameters.

Scientific impact: Medium — interesting but harder to validate without in situ data at multiple depths.

Temperature-dependent emissivity

Standard models use constant $\epsilon = 0.95$. But emissivity of silicates varies with T : at $T < 100$ K (PSRs), emissivity may drop to ~ 0.9 or lower due to reduced phonon absorption. This directly affects the surface energy balance.

Implementation difficulty: Low. Just modify the boundary condition by a few lines.

Scientific impact: Medium — small effect but physically motivated and untested for PSRs.

Vapor diffusion term in the heat equation

Add a latent heat source/sink term: $\rho \cdot c_p \cdot \partial T / \partial t = \partial / \partial z [K \cdot \partial T / \partial z] + L \cdot \partial \phi_{\text{ice}} / \partial t$, where $L = 2.83 \times 10^6$ J/kg is the latent heat of sublimation. When ice sublimates, it cools the surrounding regolith; when it recondenses, it warms it.

Implementation difficulty: High. Requires coupling a vapor transport equation to the heat equation.

Scientific impact: High — genuinely new but may be too ambitious for the timeline.

Rock abundance in the thermal model

At 20 m resolution, individual rocks are resolved. Rocks have $K \sim 2$ W/m/K vs regolith $K \sim 10^{-3}$. A two-component pixel model (rock fraction from Diviner + regolith) would improve thermal predictions near crater ejecta.

Implementation difficulty: Moderate. Need Diviner rock abundance maps as input.

Scientific impact: Medium — practical for Artemis site characterization.

Recommendation: Go with the **ice-coupled thermal properties** as your primary novel contribution. It's the most impactful, directly relevant to TSUKIMI, feasible within your timeline (~50 lines of iteration wrapper), and produces a result nobody has published. The temperature-dependent emissivity is a nice secondary addition if you have time — it's only a few lines and physically motivated for PSR work.

5. Claude Code Skill: How to Use It

A custom SKILL.md file has been created that turns Claude Code into a domain expert for your project. Here's how to set it up:

Setup steps:

1. In your Lunar-Clean repository, create a directory:
`.claude/skills/`
2. Place the SKILL.md file there:
`.claude/skills/lunar-thermal.md`
3. Alternatively, if using Claude Code's project-level instructions, paste the content into your CLAUDE.md file at the repository root.

What the skill does:

- Enforces correct physics (H-parameter, geometric grid, proper BCs)
- Catches known bugs (bottom BC, k_d value, spin-up cycles)
- Flags unsourced numerical claims
- Remembers project decisions (discrete model retired, Diviner is primary validation, etc.)
- Knows the full repository structure and where each module lives
- Understands the novel ice-coupled contribution and protects it
- Formats output for TSUKIMI RTM compatibility